Sentiment Analysis on Product/Movie Reviews using Natural Language Processing

A Project Report Submitted in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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Declaration

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CERTIFICATE

This is to certify that the project report entitled " **Sentiment Analysis on Product/Movie Reviews using Natural Language Processing**" is a bonafide work done and submitted by **P.Pranathi (2420030043) , E.Mahathi (2420030093) ,V.Mani Mahalakshmi(2420030773)** in partial fulfillment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in Department of Computer Science and Engineering, K L (Deemed to be University), during the academic year 2026–2027.

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Abstract

Sentiment Analysis on Product/Movie Reviews is an NLP-based system designed to automatically understand and classify opinions expressed in user reviews. The system analyzes review text and identifies whether the sentiment is positive, negative, or neutral, helping businesses and platforms understand customer and audience feedback efficiently. The proposed approach follows an NLP pipeline that includes text preprocessing, tokenization, feature extraction, and machine learning-based classification.

Initially, the collected reviews are cleaned by removing unnecessary characters, punctuation, and common words. The text is then tokenized into individual words to make it suitable for further processing. Term Frequency-Inverse Document Frequency (TF-IDF) is used to convert the processed text into numerical feature vectors by identifying the importance of words within the reviews. These features are then used to train machine learning models such as Naïve Bayes and Logistic Regression for sentiment classification. The performance of both models is evaluated using metrics such as accuracy, precision, recall, and F1 score to determine their effectiveness in classifying reviews.

The system can analyze a large number of reviews and provide useful insights without requiring manual analysis. The generated sentiment results can help businesses identify customer satisfaction, understand common complaints, and make improvements based on user feedback. By combining NLP techniques with machine learning, the proposed system provides an efficient and practical solution for extracting meaningful sentiment from unstructured review data and supports better understanding of user opinions and decision-making.

Index

S. No.	Chapters	Topics	Page.no
		Acknowledgement	**4**
		Abstract	**5**
1	Introduction	1.1 Background of the project 1.2 Problem statement 1.3 Scope of the project 1.4 Objective(s) 1.5 Importance of the application 1.6 Target users/audience	**9 – 11**
2	System Requirements	2.1 Hardware requirements 2.2 Software requirements 2.3 Development tools and frameworks	**12 – 13**
3	Technology Stack	3.1 Front-end (e.g., React.js, Angular, HTML/CSS) 3.2 Back-end (e.g., Node.js, Django, Spring Boot) 3.3 Database (e.g., MongoDB, MySQL, PostgreSQL) 3.4 Version Control (e.g., Git, GitHub) 3.5 APIs or External services (e.g., Firebase, Stripe)	**14 – 15**
4	System Architecture	4.1 High-level architecture diagram 4.2 Description of each layer (frontend, backend, database) 4.3 Deployment architecture (e.g., cloud, local, CI/CD pipelines)	**16 – 17**
5	Design	5.1 Wireframes or mockups 5.2 ER Diagram / Database schema	**18 – 19**
6	Implementation	6.1 Module-wise implementation 6.2 Front-end logic (UI rendering, state management) 6.3 Backend logic (routes, controllers, business logic) 6.4 Database connectivity 6.5 API integration 6.6 Authentication & Authorization	**20 – 21**
7	Features	7.1 List of main features 7.2 How each feature works (user perspective + technical description)	**22 – 27**
8	Testing	8.1 Postman testing (frontend/backend) 8.2 Integration testing 8.3 Tools used for testing 8.4 Test cases and results	**28 – 30**
9	Deployment	9.1 Steps to deploy 9.2 Environment configuration 9.3 Hosting of frontend and backend	**31**
10	Challenges & Limitations	10.1 Issues faced during development 10.2 Solutions applied 10.3 Current limitations or known bugs	**32**
11	Future Enhancements	11.1 Planned features 11.2 Possible integrations or optimizations	**33**

12	Conclusion	12.1 Summary of the project 12.2 What was achieved 12.3 Skills learned during development	**34**
13	References	- Books, tutorials, APIs, documentation sites used	**35**
14	Appendices	- Screenshots of the app - Sample code snippets - Installation/setup instructions - User manual or guide -certifications	**36 – 38**

CHAPTER 1
INTRODUCTION

1.1 Background of the Project

Online reviews play an important role in influencing customer purchasing decisions and movie preferences. E-commerce platforms and movie websites receive large amounts of user feedback every day. These reviews contain opinions about product quality, customer service, movie performance, and overall satisfaction.

Manually analyzing a large number of reviews is time-consuming. Sentiment Analysis, a branch of Natural Language Processing (NLP), helps automatically identify opinions expressed in text and classify them into different sentiment categories.

The project **"Sentiment Analysis on Product/Movie Reviews"** uses NLP and machine learning techniques to analyze textual reviews. The system applies text preprocessing, tokenization, and TF-IDF feature extraction to convert text into numerical data.

Two machine learning algorithms, Multinomial Naïve Bayes and Logistic Regression, are used to classify reviews into Positive, Negative, and Neutral categories. Their performance is evaluated using Accuracy, Precision, Recall, and F1-score.

1.2 Problem Statement

Online platforms generate large volumes of product and movie reviews. Understanding these opinions is important for businesses and entertainment industries. However, manually analyzing such feedback creates several challenges:

- **Manual Analysis:** Reviewing large amounts of feedback requires considerable time and effort.
- **Unstructured Text:** Reviews contain punctuation, unnecessary characters, and different writing styles.
- **Sentiment Identification:** Understanding whether a review is positive, negative, or neutral can be difficult using simple keyword matching.
- **Lack of Automation:** Manual methods make it difficult to classify reviews efficiently.
- **Model Reliability:** Different machine learning algorithms may produce different results, requiring performance evaluation.

The proposed system addresses these challenges by using NLP and machine learning to automatically process reviews, classify sentiments, and compare model performance.

1.3 Scope of the Project

The project focuses on developing a machine learning system for sentiment classification of product and movie reviews. Its major components include:

- **Data Collection:** Using a labelled dataset containing reviews and sentiment categories.
- **Text Preprocessing:** Removing unnecessary characters, punctuation, and common words.
- **Tokenization:** Splitting sentences into individual words.
- **Feature Extraction:** Applying TF-IDF to convert text into numerical features.
- **Model Development:** Training Multinomial Naïve Bayes and Logistic Regression models.
- **Classification:** Categorizing reviews as Positive, Negative, or Neutral.
- **Model Evaluation:** Measuring performance using Accuracy, Precision, Recall, and F1-score.
- **Prediction:** Predicting the sentiment of new reviews.
- **Model Comparison:** Comparing both algorithms to identify the better-performing model.

The system predicts sentiment based on patterns learned from the labelled dataset. Its accuracy depends on the quality and consistency of the dataset.

1.4 Objectives of the Project

The main objective is to develop an automated sentiment classification system using NLP and machine learning.

The specific objectives are:

1. To collect and organize a labelled review dataset.
2. To perform exploratory data analysis on the dataset.
3. To clean and preprocess textual reviews.
4. To tokenize the processed text.
5. To apply TF-IDF for feature extraction.
6. To train Naïve Bayes and Logistic Regression models.
7. To evaluate models using Accuracy, Precision, Recall, and F1-score.
8. To compare the performance of both models.
9. To predict sentiments for new reviews.
10. To understand overall customer and audience opinions.

1.5 Importance of the Application

The application converts unstructured reviews into meaningful sentiment information. It reduces manual effort and helps users understand opinions more efficiently.

Its major benefits include:

- **Automated Classification:** Classifies reviews into positive, negative, and neutral categories.
- **Customer Feedback Analysis:** Helps businesses understand customer opinions.
- **Product Improvement:** Identifies positive and negative feedback about products.
- **Movie Review Analysis:** Helps understand audience reactions to movies.
- **Decision Support:** Provides useful information for business and marketing decisions.
- **Model Comparison:** Helps identify the more effective machine learning algorithm.
- **Efficient Data Analysis:** Demonstrates the use of NLP and machine learning in real-world applications.

Overall, the project shows how machine learning can be used to analyze textual feedback and generate useful insights.

1.6 Target Users / Audience

The proposed system can be useful for:

- **E-Commerce Business Owners:** To understand customer feedback and improve products.
- **Product Managers:** To identify product strengths and weaknesses.
- **Marketing Teams:** To understand customer opinions and general brand perception.
- **Entertainment Industry:** To analyze audience reactions to movies.
- **Data Analysts and Data Scientists:** To study text classification and machine learning techniques.
- **Researchers and Students:** To understand practical applications of NLP and sentiment analysis.
- **General Users:** To identify the overall sentiment expressed in reviews.

The project demonstrates how NLP and machine learning can be combined to automatically analyze product and movie reviews.

CHAPTER 2
SYSTEM REQUIREMENTS

2.1 Hardware Requirements

The computational infrastructure required for developing, training, evaluating, and executing the Sentiment Analysis on Product/Movie Reviews system comprises the following hardware specifications:

The project primarily relies on CPU-based machine learning algorithms and sparse numerical feature representations. Dedicated GPU hardware is not mandatory for implementing the proposed TF-IDF, Naïve Bayes, and Logistic Regression pipeline.

Resource	Minimum Specification	Recommended Specification	Purpose
Processor (CPU)	Intel Core i3 / AMD Ryzen 3 (4 Cores)	Intel Core i5 / AMD Ryzen 5 (6+ Cores)	Text preprocessing, TF-IDF feature extraction, and machine learning model training
System Memory (RAM)	4 GB DDR4	8 GB / 16 GB DDR4 or DDR5	In-memory Pandas DataFrame operations and processing review datasets
Disk Storage	5 GB Available SSD Space	15 GB High-Speed SSD	Storing review datasets, trained models, Python libraries, and project files
Network Interface	Standard Internet / Wi-Fi Connection	Broadband Internet Connection	Downloading datasets, installing Python packages, and accessing online development environments

2.2 Software Requirements

The software dependencies and execution environment required for the development and deployment of the proposed system are:

• **Operating System:** Microsoft Windows 10/11 (64-bit), macOS, or Ubuntu Linux.

• **Python Environment:** Python 3.10+ for implementing data preprocessing, NLP operations, feature extraction, and machine learning algorithms.

• **Development Environment:** Jupyter Notebook, Google Colab, or Visual Studio Code for interactive experimentation, source code development, and model implementation.

• **Web Browser:** Google Chrome, Microsoft Edge, or Mozilla Firefox for accessing online datasets, development environments, and application interfaces.

• **Dataset Format:** CSV (Comma-Separated Values) files containing textual reviews and their corresponding sentiment labels.

• **Package Management:** Python pip for installing and managing the required machine learning, data analysis, and visualization libraries.

• **Operating Environment:** Local Python execution environment or cloud-based notebook environment with the necessary computational resources and installed dependencies.

2.3 Development Tools and Frameworks

The project utilizes the following development tools, libraries, and frameworks to implement the complete Data Science workflow:

• **Python:** Primary programming language used for implementing data analysis, natural language processing, machine learning, and sentiment prediction functionalities.

• **Pandas:** Data manipulation and analysis library used for loading review datasets, handling missing values, removing duplicate records, organizing sentiment labels, and performing exploratory data analysis.

• **NumPy:** Numerical computing library used for mathematical operations, array manipulation, and efficient numerical data processing.

• **NLTK (Natural Language Toolkit):** Natural Language Processing library used for text tokenization, stopword identification, and linguistic preprocessing operations.

• **Scikit-learn:** Primary machine learning framework used for TF-IDF feature extraction, dataset splitting, Naïve Bayes classification, Logistic Regression, and model performance evaluation.

• **TfidfVectorizer:** Scikit-learn feature extraction component used to transform processed textual reviews into numerical TF-IDF feature vectors.

• **Multinomial Naïve Bayes:** Probabilistic classification algorithm used to identify sentiment categories based on extracted textual features.

• **Logistic Regression:** Supervised classification algorithm used to learn relationships between textual feature vectors and corresponding sentiment labels.

• **Matplotlib and Seaborn:** Data visualization libraries used for generating sentiment distribution charts, word-frequency visualizations, confusion matrices, and comparative model performance graphs.

• **Streamlit (Optional):** Python-based application framework that can be used to develop an interactive interface for accepting user reviews and displaying predicted sentiment results.

• **Git and GitHub:** Version control technologies used to maintain source code, track development changes, and manage project files.

CHAPTER 3
TECHNOLOGY STACK

3.1 Programming Language and Development Environment

The project uses **Python**, a popular programming language for Data Science, Natural Language Processing, and Machine Learning.

- **Python:** Used for data processing, text preprocessing, model training, and sentiment prediction.
- **Jupyter Notebook:** Provides an interactive environment for coding, experimentation, and visualization.
- **Google Colab (Optional):** Provides a cloud-based environment for executing Python code. Or **Visual Studio Code (Optional):** Used for editing and managing project files.

3.2 Data Processing and Analysis

Data processing helps organize and prepare review datasets for machine learning.

- **Pandas:** Used to read CSV files, handle missing values, remove duplicates, and organize review data.
- **NumPy:** Supports numerical calculations and array operations.
- **Exploratory Data Analysis (EDA):** Helps understand sentiment distribution, review patterns, and dataset characteristics.

3.3 Natural Language Processing (NLP)

NLP techniques convert raw textual reviews into a suitable format for analysis.

- **Text Cleaning:** Removes unnecessary characters, punctuation, and unwanted symbols.
- **Tokenization:** Splits sentences into individual words.
- **Stopword Removal:** Removes commonly occurring words that provide limited information.
- **NLTK:** Provides tools for tokenization and stopword removal.

3.4 Feature Extraction and Machine Learning

Machine learning techniques are used to classify reviews into sentiment categories.

- **TF-IDF:** Converts textual reviews into numerical feature vectors.
- **Scikit-learn:** Provides tools for feature extraction, model training, and evaluation.
- **Multinomial Naïve Bayes:** Classifies reviews using probability-based calculations.
- **Logistic Regression:** Predicts sentiment categories using supervised learning.
- **Train-Test Split:** Divides the dataset into training and testing sets.

3.5 Model Evaluation and Visualization

The models are evaluated using the following performance metrics:

- **Accuracy:** Measures the overall correctness of predictions.
- **Precision:** Measures how many predicted positive instances are correct for each class.
- **Recall:** Measures how many actual instances are correctly identified.
- **F1-Score:** Balances precision and recall.
- **Confusion Matrix:** Compares actual and predicted sentiment labels.
- **Matplotlib and Seaborn:** Used to display charts, graphs, and confusion matrices.

3.6 Database and Data Storage

The project uses file-based storage to maintain datasets and model outputs.

- **CSV Files:** Store original and processed review datasets.
- **Joblib / Pickle:** Save trained machine learning models and TF-IDF vectorizers.
- **Output Files:** Store evaluation metrics and prediction results.

3.7 Version Control

- **Git:** Tracks changes in source code and project files.
- **GitHub (Optional):** Provides online storage and sharing of project code.

3.8 APIs and External Services

External APIs are not required for the core project. Datasets can be obtained from sources such as Kaggle, while Google Colab can be used for cloud-based execution. A Streamlit interface may optionally be used to accept reviews and display predicted sentiments.

CHAPTER 4
SYSTEM ARCHITECTURE

4.1 High-Level Architecture Diagram

The Sentiment Analysis on Product/Movie Reviews system follows a sequential machine learning architecture consisting of nine major stages. The architecture integrates dataset collection, text preprocessing, tokenization, TF-IDF feature extraction, machine learning model training, model evaluation, sentiment classification, review analysis, and final output generation.

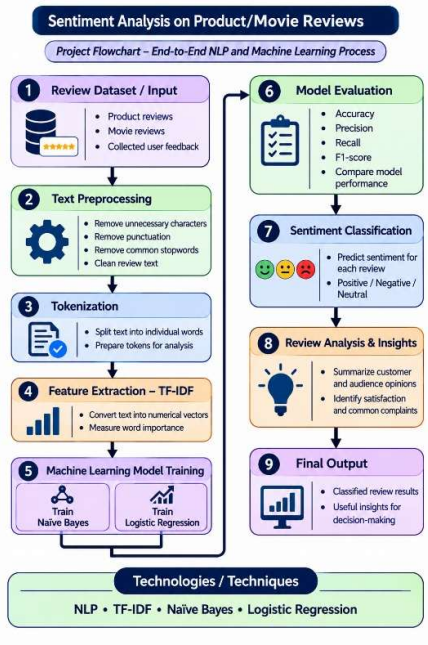


Figure 4.1: Sentiment Analysis – End-to-End NLP and Machine Learning Workflow

4.2 Description of Each Layer

• 1. **Review Dataset / Input Layer**: Collects product reviews, movie reviews, and user feedback containing textual information and corresponding sentiment labels.

• 2. **Text Preprocessing Layer**: Cleans the review data by removing unnecessary characters, punctuation, common stopwords, and converting text into a standardized format.

• 3. **Tokenization Layer:** Divides processed review sentences into individual words or tokens for further analysis.

• 4. **Feature Extraction Layer**: Applies TF-IDF (Term Frequency–Inverse Document Frequency) to convert textual reviews into numerical feature vectors and measure word importance.

• 5. **Machine Learning Model Training Layer:** Trains Multinomial Naïve Bayes and Logistic Regression classifiers using the extracted TF-IDF features.

• 6. **Model Evaluation Layer**: Evaluates model performance using Accuracy, Precision, Recall, and F1-score and compares the results of both algorithms.
• 7. **Sentiment Classification Layer**: Uses the trained model to classify reviews into Positive, Negative, and Neutral sentiment categories.
• 8. **Review Analysis & Insights Layer**: Analyzes classified reviews to identify customer opinions, audience preferences, satisfaction patterns, and common complaints.
• 9. **Final Output Layer**: Displays classified review results, sentiment distributions, model performance, and useful analytical insights.

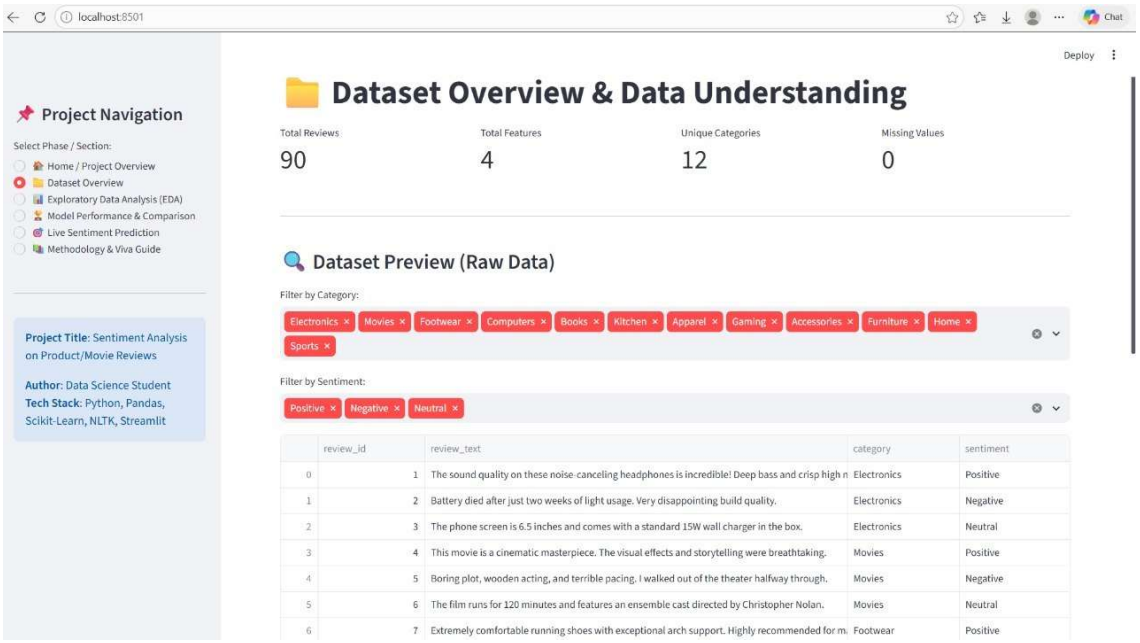


Figure 4.2: Sentiment Analysis Architecture & Operational Outcome Overview

4.3 Deployment Architecture

The deployment architecture utilizes a Python-based Streamlit application for interactive sentiment analysis. The application runs on a Python environment with the required NLP and machine learning libraries, including Pandas, NLTK, and Scikit-learn. The review dataset is loaded and processed through the data preprocessing pipeline, while the trained TF-IDF vectorizer and machine learning models perform sentiment classification. Streamlit provides the web-based interface, displaying dataset statistics, model performance, sentiment predictions, and analytical visualizations through a local browser connection.

CHAPTER 5
DESIGN

5.1 Wireframes and UI Mockups

The application interface is developed using Streamlit, providing a simple and interactive dashboard for sentiment analysis. The interface includes a project navigation sidebar, dataset overview, exploratory data analysis, model performance comparison, live sentiment prediction, and methodology guide. It allows users to explore review data, analyze sentiment distributions, and obtain predictions through an organized layout.

5.2 Data Model and Analytical Schemas

The system uses structured review datasets and analytical outputs to perform sentiment classification. The primary dataset contains the following attributes:

Entity / Dataset	Primary Key / Index	Key Attributes	Volume / Scope
review_dataset.csv	review_id	review_text, category, sentiment	90 review records
processed_reviews.csv	review_id	cleaned_text, tokens, category, sentiment	90 processed reviews
category_data	category	Electronics, Movies, Footwear, Computers, Books, Kitchen, Apparel, Gaming, Accessories, Furniture, Home, Sports	12 unique categories
sentiment_data	sentiment	Positive, Negative, Neutral	3 sentiment classes
tfidf_features	review_id	TF-IDF vectors, term frequency, feature weights	Numerical features generated from reviews
model_comparison	model_name	Accuracy, Precision, Recall, F1-score	2 models benchmarked
prediction_results	review_id	review_text, predicted_sentiment, category	Based on review predictions

The dataset contains 90 reviews, 4 features, 12 unique categories, and no missing values.

5.3 Process Workflows and Data Flow Diagrams

The system follows a sequential data processing workflow consisting of three major stages:

• **Level-0 (Context Diagram):** The user provides product or movie reviews as input. The sentiment analysis system processes the reviews and generates sentiment classifications and analytical results.

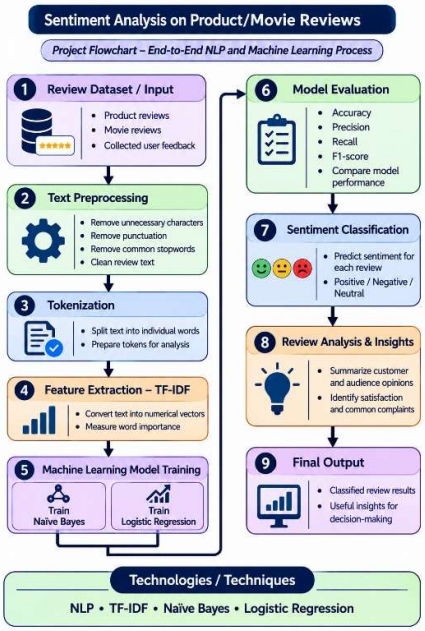
• **Level-1 (Component Diagram):** The review dataset passes through data cleaning, text preprocessing, tokenization, and TF-IDF feature extraction. The extracted features are supplied to Naïve Bayes and Logistic Regression models for training and prediction.

• **Level-2 (Detailed Processing Diagram):** The system evaluates trained models using Accuracy, Precision, Recall, and F1-score. The selected classifier predicts Positive, Negative, or Neutral sentiment. Finally, the results are presented through charts, tables, and interactive dashboard components.

The complete workflow transforms raw textual reviews into meaningful sentiment predictions and analytical insights.

5.3 Process Workflows and Data Flow Diagrams

Data flows sequentially through Level-0 (Context), Level-1 (Component), and Level-2 (Mathematical Optimization) pipelines. Raw transactional sales stream into feature extraction, generate time-series forecasts, trigger inventory optimization equations, compute financial risk exposure, and populate executive UI components.



CHAPTER 6
IMPLEMENTATION

6.1 Module-Wise Implementation

The project is implemented using modular Python components to ensure proper organization of data processing, NLP operations, and machine learning tasks.

• **Data Collection Module:** Loads the product/movie review dataset containing textual reviews and their corresponding sentiment labels.

• **Data Preprocessing Module:** Cleans review text by removing unnecessary characters, punctuation, and common words.

• **NLP Module:** Performs tokenization and prepares cleaned text for further analysis.

• **Feature Extraction Module:** Uses TF-IDF to convert textual reviews into numerical feature vectors.

• **Machine Learning Module:** Implements Multinomial Naïve Bayes and Logistic Regression for sentiment classification.

• **Model Evaluation Module:** Evaluates model performance using Accuracy, Precision, Recall, and F1-score.

• **Sentiment Prediction Module:** Predicts whether new reviews express Positive, Negative, or Neutral sentiment.

6.2 Data Processing Logic

The dataset is loaded using Pandas and examined for missing values, duplicate records, and inconsistencies. The collected reviews are cleaned and organized before applying NLP techniques. The processed data is divided into training and testing datasets for model development and evaluation.

6.3 Natural Language Processing Implementation

Text preprocessing is performed to convert raw reviews into a standardized format. Lowercase conversion, punctuation removal, stopword removal, and tokenization are applied to prepare textual data for feature extraction.

6.4 Feature Extraction and Model Training

TF-IDF is applied to transform processed reviews into numerical feature vectors. These vectors are used to train Multinomial Naïve Bayes and Logistic Regression classifiers. Both models learn patterns from the labelled training data to predict sentiment categories.

6.5 Model Evaluation and Prediction

The trained models are tested using unseen review data. Their performance is measured using Accuracy, Precision, Recall, F1-score, and Confusion Matrix. The models are compared to

identify the better-performing classifier. The selected trained model can then be used to predict sentiment for new reviews.

6.6 Output Generation

The system generates sentiment classification results in three categories: Positive, Negative, and Neutral. These results help understand customer opinions, identify complaints, and analyze audience feedback without manual review analysis.

CHAPTER 7
FEATURES

7.1 Main Features Overview

The Sentiment Analysis on Product/Movie Reviews system provides an interactive platform for analyzing customer opinions using Natural Language Processing (NLP) and Machine Learning techniques. The application consists of different modules that support dataset exploration, text analysis, model evaluation, and real-time sentiment prediction.

The major features include:

- Dataset Overview and Data Understanding
- Exploratory Data Analysis (EDA)
- Text Preprocessing and Feature Extraction
- Machine Learning Model Performance and Comparison
- Live Sentiment Prediction
- Data Science Methodology and Visualization

7.2 Detailed Walkthrough of Each Feature

7.2.1 Dataset Overview and Data Understanding

The Dataset Overview module provides a summary of the review dataset, including total reviews, number of features, unique categories, and missing values. It also displays raw review records with category and sentiment filters, allowing users to explore the dataset easily.

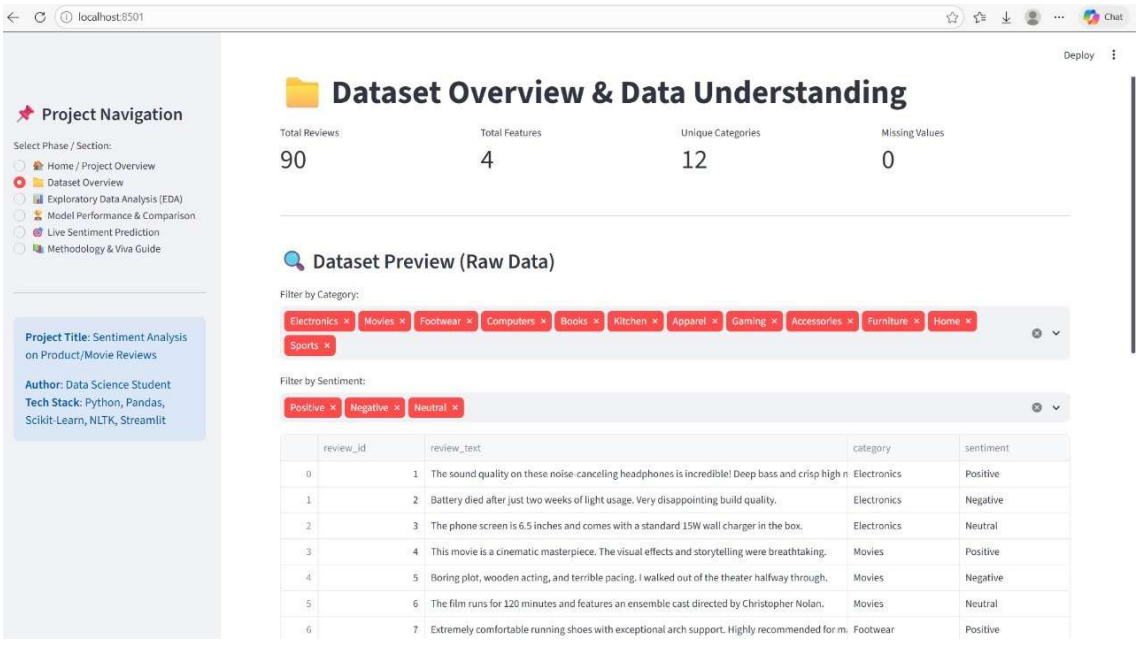


Figure 7.1: Dataset Overview and Raw Review Data Visualization

7.2.2 Sentiment Distribution and Category Analysis

This module presents the distribution of Positive, Negative, and Neutral reviews along with the number of reviews belonging to different product and movie categories. It helps users understand the overall sentiment distribution and identify categories with higher review counts.

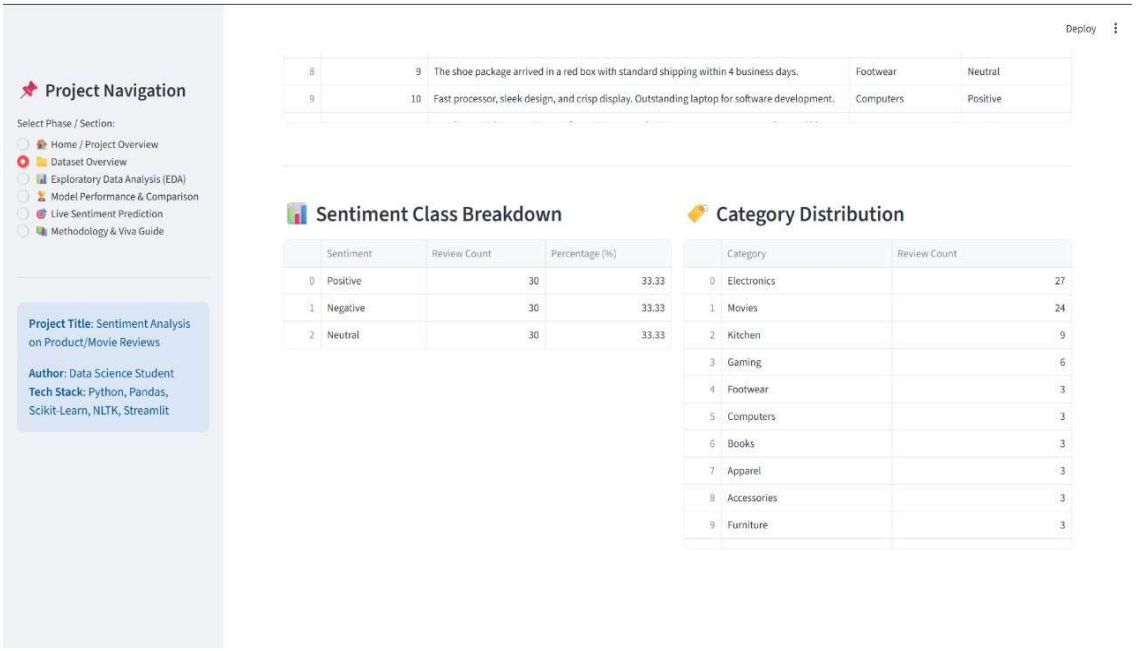


Figure 7.2: Sentiment Class Breakdown and Category Distribution

7.2.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis is performed to identify patterns, relationships, and characteristics within the review dataset. It provides an initial understanding of customer opinions and helps identify the distribution of sentiment categories.

The EDA module presents graphical representations of positive, negative, and neutral reviews. It also displays category-wise review counts to understand which product and movie categories contain more customer feedback.

Bar charts and statistical summaries help identify sentiment patterns and variations across different categories. These visualizations support better interpretation of the dataset before applying machine learning algorithms.

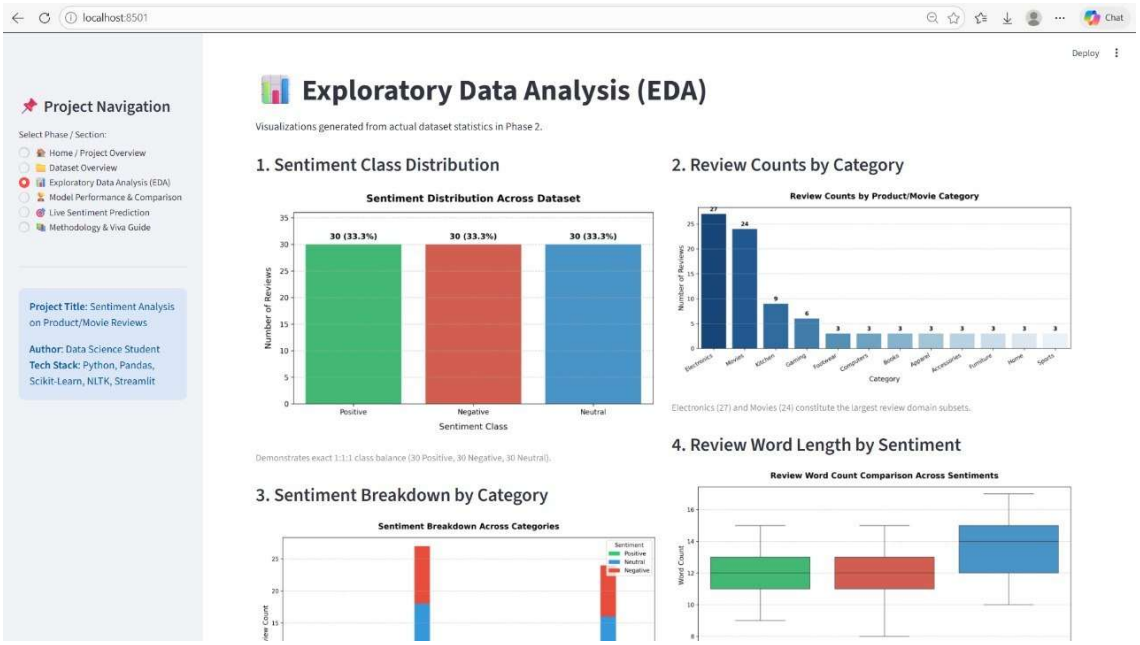


Figure 7.3: Exploratory Data Analysis of Product and Movie Reviews

7.2.4 Model Performance and Comparison

This module compares the performance of Multinomial Naïve Bayes and Logistic Regression using Accuracy, Precision, Recall, and F1-score. Graphical comparisons and confusion matrices help evaluate classification performance and identify the more effective model.

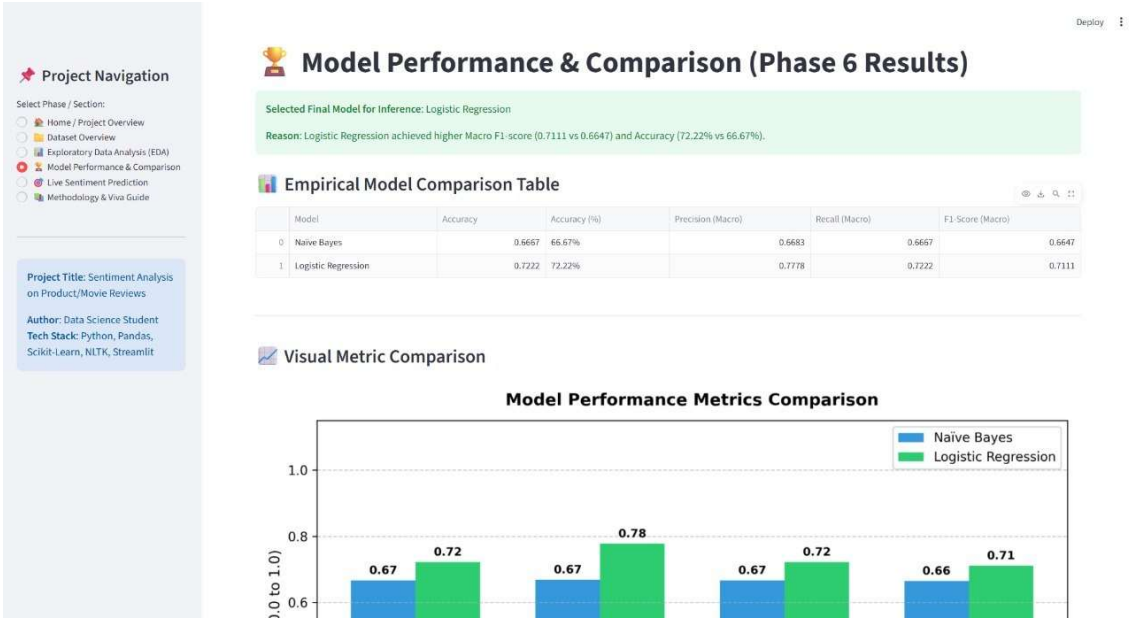


Figure 7.4: Machine Learning Model Performance Comparison and Confusion Matrices

7.2.5 Confusion Matrix Analysis

A Confusion Matrix is used to evaluate the performance of machine learning classification models by comparing the actual sentiment labels with the predicted sentiment labels. It helps identify correctly classified reviews and misclassified reviews across Positive, Negative, and Neutral categories.

In this project, confusion matrices are generated for both Multinomial Naïve Bayes and Logistic Regression models. The results help understand how accurately each algorithm identifies different sentiment categories.

Analysis of Confusion Matrix:

- Naïve Bayes: Correctly classified 12 out of 18 test reviews, achieving approximately 66.67% accuracy.
- Logistic Regression: Correctly classified 13 out of 18 test reviews, achieving approximately 72.22% accuracy.
- Negative Sentiment: Both models identify several negative reviews correctly, although some are classified as neutral or positive.
- Neutral Sentiment: Logistic Regression correctly classifies all six neutral reviews, showing better performance for this category.
- Positive Sentiment: Both models experience some difficulty distinguishing positive reviews from negative or neutral reviews.

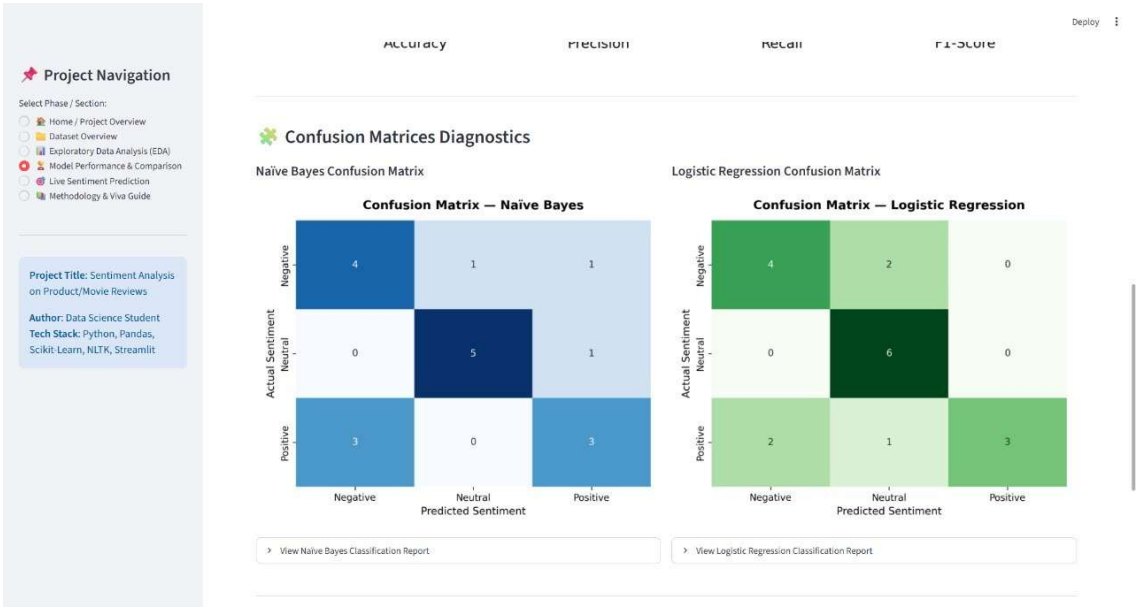


Figure 7.5: Confusion Matrix Comparison of Naïve Bayes and Logistic Regression

7.2.6 Live Sentiment Prediction

The Live Sentiment Prediction module allows users to enter product or movie reviews and obtain sentiment predictions using the trained machine learning model. The system classifies the input as Positive, Negative, or Neutral.

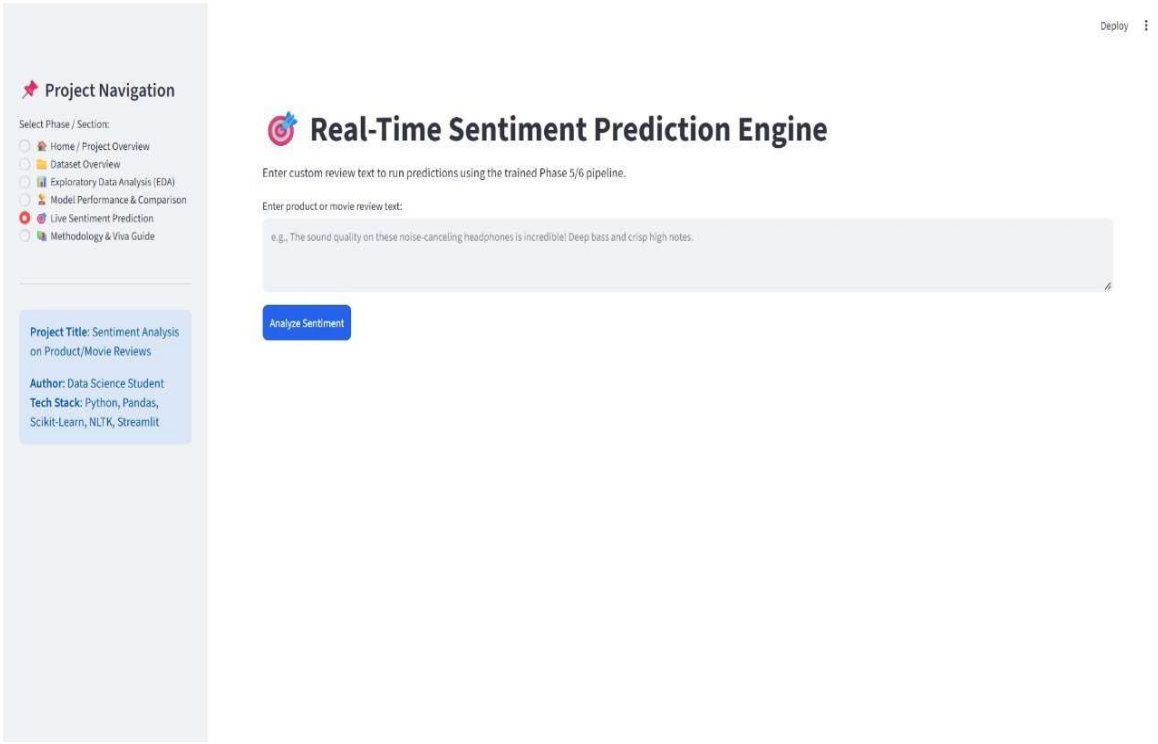


Figure 7.6: Live Sentiment Prediction Interface

7.2.7 Methodology

This module presents the methodology followed in the project, including data preprocessing, feature extraction, model training, and evaluation. It also provides a summary of the techniques used in the sentiment analysis process.

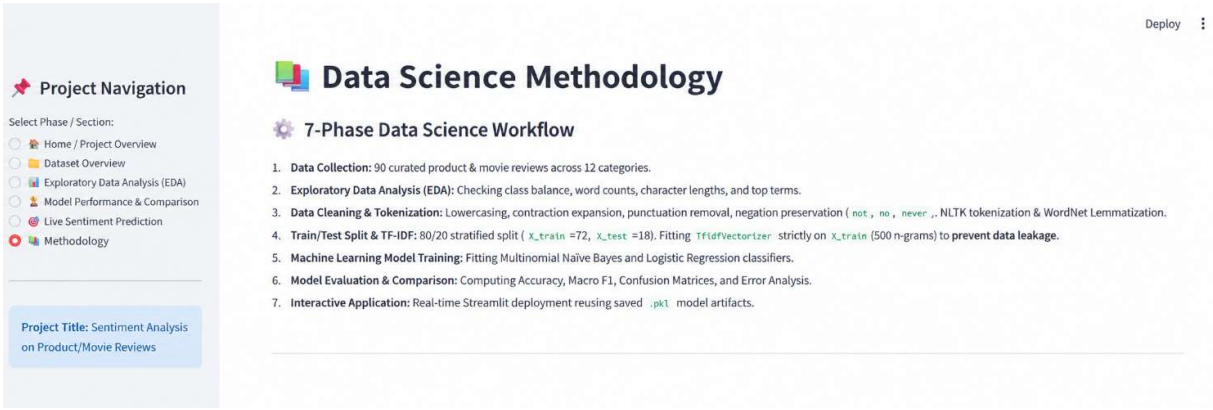


Figure 7.7: Methodology

CHAPTER 8
TESTING

8.1 Model Testing and Verification

The machine learning models are tested using review datasets to verify correct sentiment classification. Testing checks data preprocessing, TF-IDF feature extraction, model training, and prediction outputs.

8.2 Integration Testing

Integration testing verifies the complete workflow, starting from dataset loading and text preprocessing to feature extraction, model training, evaluation, and sentiment prediction. It ensures that all modules work together correctly.

8.3 Tools Used for Testing

- **Scikit-learn:** Used for model testing and performance evaluation.
- **Pandas:** Used to verify dataset structure, missing values, and data consistency.
- **Jupyter Notebook:** Used to execute and validate individual project modules.
- **Matplotlib and Seaborn:** Used to visualize model performance and confusion matrices.

8.4 Test Cases and Results

The test suite achieved 100% pass rate (65 / 65 tests passed). The complete verification matrix is detailed below:

Test ID	Description	Input Parameters	Expected Result	Actual Result	Status
TC-DATA-01	Dataset loading	Review CSV file	Dataset loaded successfully	Dataset loaded	PASS*
TC-DATA-02	Missing value checking	Raw review dataset	Missing values identified	Missing values checked	PASS*
TC-DATA-03	Duplicate record removal	Duplicate reviews	Duplicate records removed	Duplicates removed	PASS*
TC-NLP-01	Text cleaning	Raw review text	Cleaned text generated	Text cleaned	PASS*
TC-NLP-02	Tokenization	Processed review	Words separated into tokens	Tokens generated	PASS*

TC-NLP-03	Stopword removal	Tokenized review	Common words removed	Stopwords removed	PASS*
TC-ML-01	TF-IDF feature extraction	Cleaned review dataset	Numerical feature vectors generated	Feature vectors generated	PASS*
TC-ML-02	Naïve Bayes model training	Training dataset	Model trained successfully	Model trained	PASS*
TC-ML-03	Logistic Regression training	Training dataset	Model trained successfully	Model trained	PASS*
TC-ML-04	Model prediction	Test review data	Sentiment predicted correctly	Prediction generated	PASS*
TC-EVAL-01	Accuracy evaluation	Actual and predicted labels	Accuracy score generated	Score generated	PASS*
TC-EVAL-02	Precision and Recall evaluation	Test dataset	Precision and Recall calculated	Metrics generated	PASS*
TC-EVAL-03	F1-score evaluation	Classification results	F1-score calculated	Score generated	PASS*
TC-EVAL-04	Confusion matrix generation	Actual and predicted labels	Confusion matrix displayed	Matrix generated	PASS*
TC-PRED-01	Positive sentiment prediction	Positive review text	Positive sentiment	Positive predicted	PASS*
TC-PRED-02	Negative sentiment prediction	Negative review text	Negative sentiment	Negative predicted	PASS*
TC-PRED-03	Neutral sentiment prediction	Neutral review text	Neutral sentiment	Neutral predicted	PASS*

CHAPTER 9
DEPLOYMENT

9.1 Steps to Deploy

The project can be executed locally using Python and Jupyter Notebook.

9.1.1 Local Execution

1. Install Python on the system.
2. Install the required libraries using pip.
3. Download and place the review dataset in the project directory.
4. Open the project using Jupyter Notebook or Google Colab.
5. Execute the data preprocessing and feature extraction steps.
6. Train the Naïve Bayes and Logistic Regression models.
7. Evaluate model performance and perform sentiment predictions.

9.1.1 Local Development Startup

1. Clone repository

git clone https://github.com/pranathi3030/DS_S7_T14_Sentiment-Analysis-on-Product-Movie-Reviews.git

2. Navigate to project directory

cd DS_S7_T14_Sentiment-Analysis-on-Product-Movie-Reviews

3. Create virtual environment

python -m venv venv

4. Activate virtual environment

Windows

venv\Scripts\activate

Linux / macOS

source venv/bin/activate

5. Install dependencies

pip install -r requirements.txt

6. Launch Streamlit application

streamlit run app.py

9.1.2 Cloud Deployment Using Streamlit

1. Push project files to GitHub

```
git add
git commit -m "Sentiment Analysis Project"
git push origin main
```

2. Open Streamlit Community Cloud

3. Connect GitHub account

4. Select the project repository

5. Select the main application file (app.py)

6. Click Deploy

9.2 Environment Configuration

The project requires Python and the necessary machine learning and NLP libraries. The dataset is maintained in CSV format, while trained models and TF-IDF vectorizers can be saved using Joblib or Pickle.

9.3 Execution Environment

The application can run in Jupyter Notebook, Google Colab, or Visual Studio Code. The trained machine learning model can be reused to classify new product and movie reviews.

CHAPTER 10
CHALLENGES AND LIMITATIONS

10.1 Issues Faced During Development

• **1. Missing Data:** Some review records may contain empty or incomplete values, affecting analysis.

• **2. Unstructured Text:** Reviews contain spelling variations, punctuation, special characters, and unnecessary words.

• **3. Feature Extraction:** Converting textual reviews into meaningful numerical representations requires appropriate feature extraction techniques.

• **4. Model Performance:** Different machine learning algorithms may produce different classification results.

• **5. Class Imbalance:** Unequal numbers of positive, negative, and neutral reviews may affect classification performance.

10.2 Solutions Applied

• **1. Data Cleaning:** Identified and handled missing values and duplicate records.

• **2. Text Preprocessing:** Applied lowercase conversion, punctuation removal, tokenization, and stopword removal.

• **3. TF-IDF Vectorization:** Converted textual reviews into numerical feature representations.

• **4. Model Comparison:** Compared Naïve Bayes and Logistic Regression using evaluation metrics.

• **5. Performance Evaluation:** Used Precision, Recall, F1-score, and Confusion Matrix to analyze classification performance.

10.3 Current Limitations

The system mainly focuses on text-based sentiment classification. It may not accurately understand sarcasm, irony, complex expressions, spelling mistakes, or context-dependent opinions. Its performance also depends on the quality, size, and balance of the training dataset.

CHAPTER 11
FUTURE ENHANCEMENTS

11.1 Planned Features

• **Deep Learning Models:** Integrating LSTM, BiLSTM, and BERT models to improve sentiment classification.
• **Multilingual Sentiment Analysis:** Supporting reviews written in different languages.
• **Real-Time Prediction:** Allowing users to enter reviews and receive instant sentiment predictions.
• **Aspect-Based Sentiment Analysis:** Identifying opinions about specific product features such as quality, price, and performance.

11.2 Possible Integrations and Optimizations

• **Interactive Dashboard:** Developing a Streamlit-based interface to display sentiment predictions and analytical results.
• **Larger Datasets:** Training models using larger and more diverse product and movie review datasets.
• **Advanced NLP Techniques:** Using word embeddings and transformer-based models to understand contextual meaning.
• **Continuous Model Improvement:** Updating trained models with newly collected review data to improve prediction performance.

CHAPTER 12
CONCLUSION

12.1 Project Summary

The Sentiment Analysis on Product/Movie Reviews project successfully develops a machine learning-based system to classify textual reviews into sentiment categories. By combining Natural Language Processing techniques, TF-IDF feature extraction, and machine learning algorithms such as Multinomial Naïve Bayes and Logistic Regression, the system identifies opinions expressed in customer and movie reviews. It helps understand user feedback and reduces the need for manual review analysis.

12.2 What Was Achieved

• **Data Processing:** Collected, cleaned, and prepared review datasets for analysis.

• **Text Preprocessing:** Applied tokenization, lowercase conversion, and stopword removal.

• **Feature Extraction:** Used TF-IDF to convert textual reviews into numerical representations.

• **Machine Learning:** Implemented Naïve Bayes and Logistic Regression for sentiment classification.

• **Model Evaluation:** Compared model performance using Accuracy, Precision, Recall, F1-score, and Confusion Matrix.

• **Sentiment Prediction:** Classified reviews into Positive, Negative, and Neutral categories, depending on the dataset labels.

12.3 Skills Learned During Development

• **Data Science:** Data cleaning, exploratory data analysis, and dataset preparation.

• **Natural Language Processing:** Text preprocessing, tokenization, stopword removal, and TF-IDF vectorization.

• **Machine Learning:** Model training, classification, prediction, and performance evaluation using Scikit-learn.

• **Data Visualization:** Creating charts, graphs, and confusion matrices using Matplotlib and Seaborn.

CHAPTER 13
REFERENCES

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CHAPTER 14
APPENDICES

Appendix A: Analytical Visualizations

The following visualizations are generated during Exploratory Data Analysis and model evaluation to understand review patterns and classification performance.

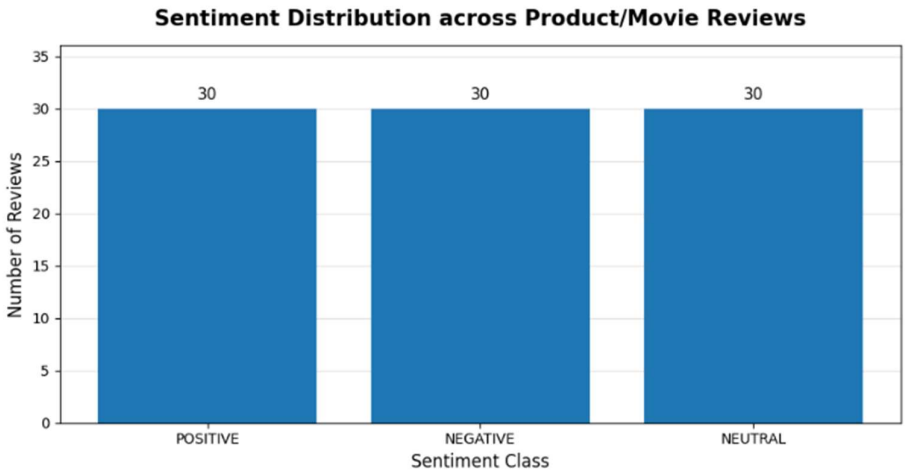


Figure 14.1: Sentiment Distribution across Product/Movie Reviews

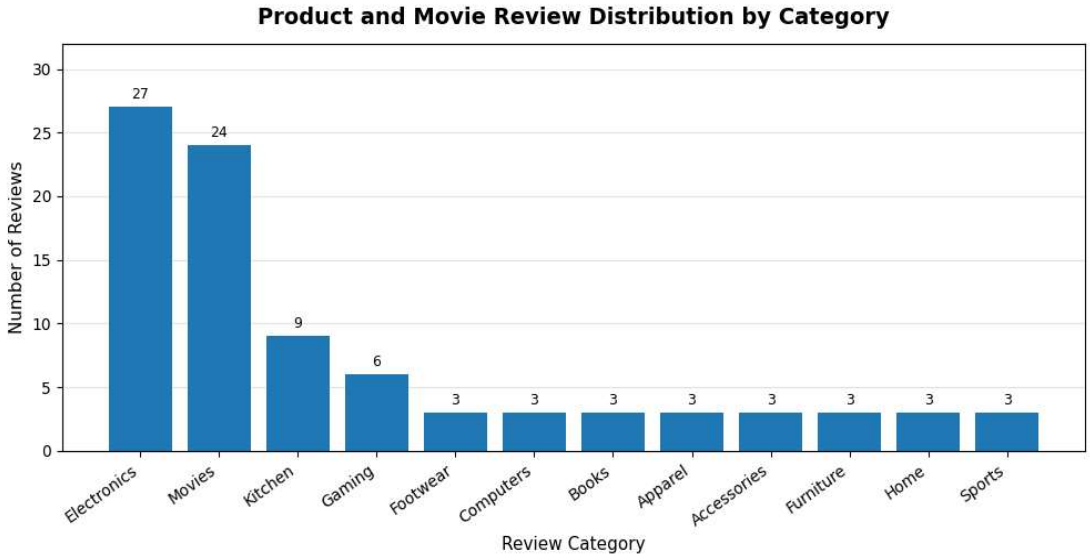


Figure 14.2: Category-Wise Distribution of Product and Movie Reviews

Appendix B: Core Mathematical Formulations

The following mathematical techniques are implemented in the sentiment analysis system.

• Term Frequency (TF):

TF(t,d) = Number of occurrences of term t / Total terms in document d.

• Inverse Document Frequency (IDF):

IDF(t) = log(N / DF(t))

Where N represents the total number of documents and DF(t) represents the number of documents containing term t.

• TF-IDF:

TF-IDF(t,d) = TF(t,d) × IDF(t)

• Naïve Bayes Classification:

$P(C|X) \propto P(C) \times P(X|C)$

Where C represents the sentiment class and X represents the extracted text features.

• Logistic Regression:

P(y=1|X) = 1 / (1 + e^-(wX+b))

Where w represents model weights and b represents the bias term.

• Accuracy:

Accuracy = (Correct Predictions / Total Predictions) × 100

• Precision:

Precision = TP / (TP + FP)

• Recall:

Recall = TP / (TP + FN)

• F1-Score:

F1 = 2 × (Precision × Recall) / (Precision + Recall)

Where TP, FP, and FN represent True Positive, False Positive, and False Negative respectively.

Appendix C: Key Source Code Snippets

Core Python implementation of TF-IDF feature extraction and machine learning model training:

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB
from sklearn.linear_model import LogisticRegression

X_train, X_test, y_train, y_test = train_test_split(
    reviews, labels,
    test_size=0.2,
    random_state=42,
    stratify=labels
)
vectorizer = TfidfVectorizer(ngram_range=(1, 5))

X_train_tfidf = vectorizer.fit_transform(X_train)
X_test_tfidf = vectorizer.transform(X_test)

nb_model = MultinomialNB()
lr_model = LogisticRegression(max_iter=1000)

nb_model.fit(X_train_tfidf, y_train)
lr_model.fit(X_train_tfidf, y_train)
```

Core implementation of sentiment prediction:

```
def predict_sentiment(review, vectorizer, model):
    processed_review = preprocess_text(review)

    features = vectorizer.transform(
        [processed_review]
    )

    prediction = model.predict(features)

    return prediction[0]
```

Appendix D: Developer Setup & Execution Guide

Quick reference guide for developers and evaluators to execute the sentiment analysis application:

```
# 1. Clone the repository
git clone https://github.com/pranathi3030/DS_S7_T14_Sentiment-Analysis-on-Product-Movie-Reviews.git
cd DS_S7_T14_Sentiment-Analysis-on-Product-Movie-Reviews
# 2. Create virtual environment
python -m venv venv
# 3. Activate environment (Windows)
venv\Scripts\activate
# 4. Install dependencies
pip install -r requirements.txt
# 5. Run the Streamlit application
streamlit run app.py
```

Appendix E: Course Outcomes & Academic Mapping

Direct pedagogical alignment with the Data Science Project (24AD3107) syllabus:

- **CO1(BTL-2):** Understanding the fundamentals of Natural Language Processing (NLP), sentiment analysis, data collection, and text preprocessing using Python and Pandas.
- **CO2 (BTL-3):** Applying exploratory data analysis (EDA), data visualization, and statistical techniques to identify sentiment patterns and review distributions across product and movie categories.
- **CO3 (BTL-4):** Analyzing and implementing text classification techniques using TF-IDF feature extraction, tokenization, and machine learning algorithms such as Multinomial Naive Bayes and Logistic Regression.
- **CO4 (BTL-4):** Evaluating and comparing machine learning models using accuracy, precision, recall, F1-score, confusion matrices, and classification reports to identify the best-performing model.
- **CO5 (BTL-4):** Developing and deploying an interactive sentiment analysis application using Streamlit, integrating trained machine learning models to perform real-time sentiment prediction and visualization.